there its gravity will take over and pull it back inward to the white-dwarf curve and a white-dwarf grave.

This instability (squeeze the $10^{13}$ star a tiny bit and it will implode to become a neutron star; expand it a tiny bit and it will explode to become a white dwarf) means that no real star can ever live for long at the $10^{13}$ point —or at any other point along the portion of the equilibrium curve marked "unstable."

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Moving on along the equilibrium curve, we meet the entire family of neutron stars, with masses ranging from 0.1 to about 2 Suns. The maximum neutron-star mass of about 2 Suns is somewhat uncertain even in the 1990s because the behavior of the nuclear force at very high densities is still not well understood. The maximum could be as low as 1.5 Suns but not much lower, or as high as 3 Suns but not much higher.

At the (approximately) 2-solar-mass peak of the equilibrium curve, the neutron stars end. As one moves further along the curve to still higher densities, the equilibrium stars become unstable in the same manner as those between white dwarfs. and neutron stars (Box 5.6). Because of this instability, these "stars," like those between white dwarfs and neutron stars, cannot exist in nature. Were they to form, they would immediately implode to become black holes or explode to become neutron stars.

Figure 5.5 is absolutely firm and unequivocal: There is *no* third family of stable, massive, cold, dead objects between the white dwarfs and the neutron stars. Therefore, when stars such as Sirius, which are more massive than about 2 Suns, exhaust their nuclear fuel, either they must eject all of their excess mass or they will implode inward past white-dwarf densities, past neutron-star densities, and into the critical circumference—where today, in the 1990s, we are completely certain they must form black holes. Implosion is compulsory. For stars of sufficiently large mass, neither the degeneracy pressure of electrons nor the nuclear force between neutrons can stop the implosion. Gravity overwhelms even the nuclear force.

There remains, however, a way out, a way to save all stars, even the most massive, from the black-hole fate: Perhaps *all* massive stars eject enough mass late in their lives (in winds or explosions), or during their deaths, to bring them below about 2 Suns so they can end up in the neutron-star or white-dwarf graveyard. During the 1940s, 1950s, and early 1960s,